B.3 Loading PDFs into Elasticsearch

Let's say our business requirement is to load PDF files (such as legal documents or medical journals in PDF format) into Elasticsearch, enabling clients to conduct searches on them. Elasticsearch lets us index PDF files using a dedicated ingest processor called `attachment`.

The `attachment` processor is used in the ingest pipeline to load attachments: PDF files, Word documents, emails, and so on. It uses the Apache Tika (https://tika.apache.org) library to extract the file data. The source data should be converted into Base64 format before being loading into the pipeline. Let's see this in action.

Continuing our MI5 example, we are expected to load all the secret data presented in PDF files into Elasticsearch. The following steps help visualize the process:

1. Define a pipeline with an `attachment` processor. The file's Base64 content is indexed into a field (for our example, we define the field as `secret_file_data`).
2. Convert the PDF file content into bytes, and feed it to the Base64 encoding utility (using any toolset at your disposal).
3. Invoke the pipeline for the incoming data so the `attachment` processor processes the data.

The following code creates a pipeline with an `attachment` processor.

Listing B.4 Pipeline with an attachment processor

```
PUT _ingest/pipeline/confidential_pdf_files_pipeline
{
  "description": "Pipeline to load PDF documents",
  "processors": [
    {
      "set": {
        "field": "category",
        "value": "confidential"
      },
      "attachment": {                          <-- The attachment processor
        "field": "secret_file_data"                with a secret_file_data field
      }
    }
  ]
}
```

When we execute this code, the `confidential_pdf_files_pipeline` is created on the cluster. The `attachment` processor expects the Base64-encoded data from a file set in the `secret_file_data` field during pipeline ingestion.

Now that we have created the pipeline, let's test it. Suppose the file data is "Sunday Lunch at Konda's" (perhaps this is code for getting rid of Mr. Konda!), and we run the Base64 encoder to produce the data in a Base64-encoded form. (I'll leave it up to you to apply an encoder—see the sidebar "Base64 encoding" for more information.)